

Figure 6: **PCFG**: An illustrative example of a PCFG and its components, with an example derivation (with final string) below. **AR vs. ATR**: Comparing AR and ATR using example documents; both tasks provide a document with key–value associations in-context and ask a query about one such association. However, associations in ATR need not involve adjacent tokens and are tree-structured.

checkpoints with non-negligible performance, the information is moved to the *next key* by the layer 0 short convolution. While moving both the key and value to the following key is unusual, it still satisfies our two-stage definition of induction, and thus we claim that Mamba too performs induction.

5 MECHANISTIC FINDINGS ON AR GENERALISE TO A NEW RETRIEVAL TASK

We now examine whether our results generalise to a novel retrieval task: **Associative Treecall** (ATR). ATR is more similar to real-world natural language retrieval than AR because it uses a probabilistic context-free grammar (PCFG) to generate hierarchical data, on which we ask AR-like queries. Since *key* and *value* need not be adjacent to each other, ATR requires a non-positional associative mechanism, which may challenge architectures that are designed for AR.

5.1 ASSOCIATIVE TREECALL (ATR)

Since a standard AR document (eq. (1)) consists of *adjacent key–value* pairs, one can associate each *key* with its corresponding *value* solely using relative position. Yet many natural language retrieval tasks require association over latent hierarchical structure. For example:

$$(2) \quad \textit{John had chicken and Mary had pork. The chicken was eaten by} \rightarrow \textit{John}$$

Answering this query requires associating *John* with *chicken* and *Mary* with *pork*, and then retrieving the appropriate association for *John*. A solution employing relative positional association would not be robust to the possible range of variation (*John had some chicken*, *John decided to have chicken*, etc.).

An ATR corpus is drawn from a synthetic probabilistic context-free grammar (PCFG) whose parameters we set. Each document consists of a string sampled from the PCFG, with latent structure made up of **parent–child** relations between symbols, followed by a divider token (EOS) and a query about one such relation. The PCFG has one special property which establishes the parent–child relationships: for the right-hand side of each production rule, the rightmost symbol is always a terminal, and is the *parent* of the symbols created by this production. We sample strings by selecting an iid nonterminal and recursively applying production rules according to the PCFG distribution. We show an example in Figure 6 and formalise definitions in §B. Since the number of tokens separating parents and their children may vary, ATR cannot be solved by a positional associative mechanism.

Setup. For each experiment, we generate a single PCFG to use across all models to ensure fair comparisons, with parameters in Table 2. We also reject any samples that have more than 1024 symbols, which only affects the sampling distribution for the most complex PCFGs we use. We follow §4.1 except we train for 32 epochs.

Each PCFG sample of length n provides us with a set of $n - 1$ eligible parent–child queries (i.e. a tree with $n - 1$ edges). However, terminals may occur multiple times, so a query about a specific symbol may present ambiguity; thus, when presenting a query we consider it to *only* refer to the

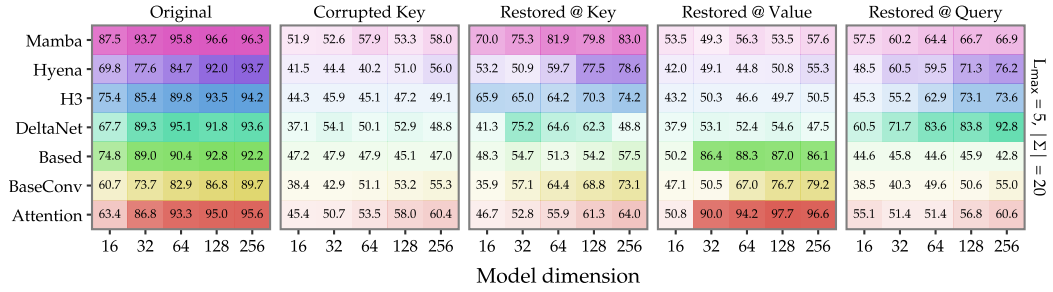


Figure 7: **Associative Treecall**: Likelihood of correct answer without any interventions (leftmost), after corrupting the key (second), and after restoring representations at the layer 1 block input (at *key*, *value*, or *query*) with interchange intervention. The pattern from AR largely holds.



Figure 8: **Ablating short convolution in Mamba on ATR**: Original, corrupted *key*, and restored interventions at layer 1 input at the *query*, *key*, and *value* in Mamba, when varying model dimensionality and conv. size. (Some checkpoints failed to train.)

rightmost instance of that symbol. To minimise the ability to heuristically guess, we inversely weight parent-child pairs by the parent’s child count when sampling queries.

5.2 PER-ARCHITECTURE MECHANISMS ARE SIMILAR BETWEEN ATR AND AR

We consider a simple setting to study models on ATR: the maximum number of symbols on the right-hand side of any rule $L_{\max} = 5$ and vocabulary size $|\Sigma| = 20$. The remaining parameters are set in accordance with §C; see Table 2 for definitions. We sweep the same model dimensionalities as in §4.2, and a smaller learning rate range of $[3 \cdot 10^{-5}, 3 \cdot 10^{-3}]$.

Behavioural results. We report results in the leftmost panel of Figure 7. Mamba is the best-performing architecture at ATR; it matches or outperforms all other architectures at all model dimensions. This is particularly surprising because longer production rules imply greater positional variation between keys and values, which ought to stress AR-optimised SSM designs like Mamba.

Unlike AR, the baseline performance on ATR after corrupting the key is well above 0. This is because some *key-value* pairings are more likely than others due to the underlying PCFG, enabling memorisation (unlike AR where all pairings are equally likely).

Mechanistic analysis. We conduct the same analysis as for AR. We recover the same overall trends but with greater inter-architecture variance: Figure 7 shows that Attention, Based, and BaseConv all primarily learn induction mechanisms with performance largely restored by *value* interventions, whereas the remaining SSMs perform direct retrieval as on AR, with performance being restored by intervening on the *key* (indicating direct retrieval by the layer 1 sequence mixer) or the *query* (indicating the same but by layer 0).

5.3 MAMBA CAN LEARN ATR WITHOUT SHORT CONVOLUTIONS

Given ATR and AR show similar patterns in mechanistic implementation across architectures, we now ask whether Mamba’s reliance on short convolution holds on this new task. Since ATR may have longer and more variable distances between the *key* and *value* than AR, short convolution (with a short receptive field that applies a fixed input-unaware transformation) may not be the appropriate operator for performing association.

Methodology. We repeat the experiments in §4.3 on ATR, training Mamba with varying conv. size or without conv. We apply mechanistic metrics to the best-performing checkpoints of each dimensionality and conv. setting.

Results. We report results in Figure 8. Unlike in AR, we find that Mamba still learns the task even without short convolutions (but not as well as with them). E.g. at $d = 256$, Mamba without short convolutions (93.4%) still outperforms Based (92.2%) and BaseConv (89.7%). The more surprising result is via mechanistic metrics: without short convolutions or with $d_{\text{conv}} = 1$, Mamba’s performance after corruption can be restored primarily by intervening at the *value*, i.e. **Mamba without short convolutions learns two-layer induction on ATR**. We have thus shown that Mamba can indeed learn an induction-like mechanism to solve associative recall tasks, albeit using different components than Attention and only on certain tasks (ATR, but not AR). This may explain other induction-like mechanisms observed in Mamba trained on natural language, e.g. Bick et al. (2025), and highlights a divergence between AR and real-world retrieval.

6 DISCUSSION

Why mechanistic evaluations over behavioural metrics? Architectural advances on language modelling are largely uncovered and presented in an empirical manner (e.g. Shazeer, 2020); beyond intuition, we have little justification as to *why* a modification or innovation improves model performance. Synthetic tasks already inform progress on architecture design (such as SSMs), but treating such tasks as just a behavioural evaluation discards useful signal; control over task parameters presents an opportunity to explain performance using interpretability.

ATR indicates induction is highly general. We introduced ATR to break the naïve key–value adjacency of AR, and see whether general mechanisms underlying association still emerge across architectures. We find the same induction mechanism, where the association is computed and stored at the value before retrieval, in Attention and Based for both tasks, as well as in Mamba without convs. While Olsson et al. (2022) and later works define induction on adjacent tokens, ATR is evidence that a *position-independent* and generalising (§D.2) notion of association is worth studying.

Short convolutions are key to association in SSMs. We showed that Mamba and Based rely on short convolutions to learn how to associate keys and values on AR and ATR. Several earlier works point to the importance of short convolution: Arora et al. (2024b) empirically show its utility on AR, Allen-Zhu & Alfarano (2025) introduce a short convolution component (Canon) in various architectures, Olsson et al. (2022) show that 1-layer attention can learn induction if augmented with a length-2 convolution; also Liu et al. (2024b); Dolga et al. (2024); Fu et al. (2023); Poli et al. (2023).

Limitations. While we proposed mechanistic evaluations as a new tool, behavioural metrics like accuracy are still needed to properly contextualise results. Additionally, we focus on synthetic tasks throughout this work; extending our analyses to real-world models would help paint a more complete picture of the differences in capabilities and mechanisms of architectures on real-world tasks.

7 CONCLUSION

In this work, we introduce mechanistic evaluations as a powerful framework for comparing model architectures. This approach goes beyond high-level behavioural metrics, revealing substantive differences between architectures. Through analysis of synthetic in-context retrieval tasks, we uncover the underlying mechanisms that explain the success and failure points of various architectures. Mechanistic evaluations thus provide a useful tool for architecture design and analysis, as well as a new opportunity for interpretability research to open the blackbox of progress in AI.